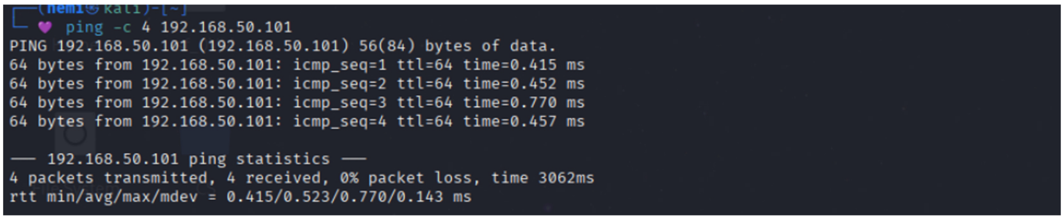
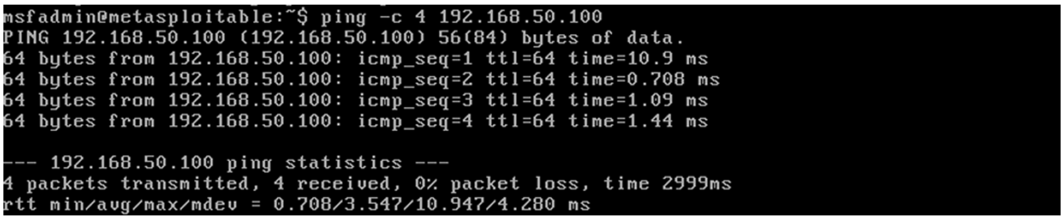
Emy Mangogna - W9D1 - Pratica

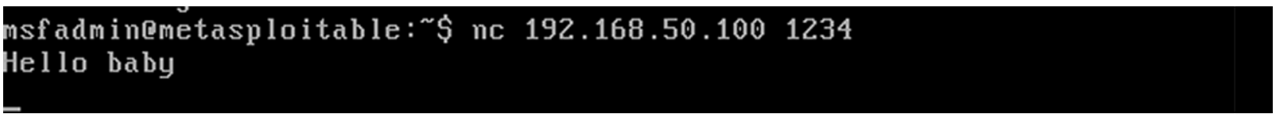
Ex 1. Utilizzo di Kali Linux (IP 192.168.50.100) come macchina attaccante per interagire con la macchina Metasploitable (IP 192.168.50.101).
Lo scopo è comprendere come stabilire connessioni client-server con **Netcat** e rilevare servizi attivi e porte aperte con **Nmap**.
Netcat è uno strumento a riga di comando per la gestione di connessioni di rete. Permette di aprire listener e stabilite connessioni client-server per l'esecuzione di comandi di base.
Nmap è uno strumento di scansione e mappatura della rete. Esegue scansioni per l'identificazione di servizi e porte aperte su un sistema target (Metasploitable).

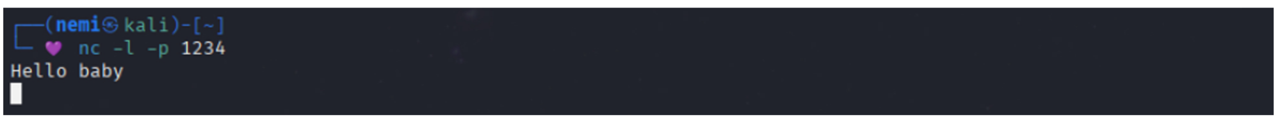
Per verificare la corretta comunicazione tra le due macchine (Kali 192.168.50.100 e Metasploitable 192.168.50.101) è stato utilizzato il comando **ping.**





Connessione client-server con Netcat:
Da kali viene lanciato il comando "**nc -l -p 1234**" , che apre un listener sulla porta **1234** per accettare connessioni in arrivo. **-l** indica l'ascolto in entrata, mentre **-p 1234** specifica la porta. Il listener è pronto a ricevere connessioni da client remoti.
Da Metasploitable è stata avviata la connessione con **nc 192.168.50.100 1234**.
Il **listener** è un programma o processo che resta in ascolto su una porta di rete specifica (1234) e attende che un client si connetta. Una volta stabilito il collegamento, è stato possibile scambiare messaggi in chiaro tra le due macchine (Hello baby).





Esecuzione di comandi remoti tramite Netcat:
Attraverso Netcat è stata dimostrata la possibilità di eseguire comandi remoti. I comandi digitati su **Kali** sono stati eseguiti e i relativi output sono stati restituiti su **Metasploitable.**

whoami mostra il nome dell'utente con cui è in esecuzione la sessione remota. Permette di identificare l'utente corrente.

```
┌──(nemi㉿kali)-[~]
└─💜 nc -l -p 1234 -c "whoami"
▌
```

```
msfadmin@metasploitable:~$ nc 192.168.50.100 1234
kali
msfadmin@metasploitable:~$ _
```

uname -a visualizza informazioni complete sul sistema operativo della macchina target: kernel, hostname, architettura e versione.

```
┌──(nemi㉿kali)-[~]
└─💜 nc -l -p 1234 -c "uname -a"
▌
```

```
msfadmin@metasploitable:~$ nc 192.168.50.100 1234
Linux kali 6.12.33+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.12.33-1kali1 (2025-0
6-25) x86_64 GNU/Linux
msfadmin@metasploitable:~$ _
```

ps -aux mostra l'elenco dei processi attivi sulla macchina target, con dettagli su PID, stato, utente e utilizzo delle risorse.

```
┌──(nemi㉿kali)-[~]
└─💜 nc -l -p 1234 -c "ps -aux"
▌
```

```
kali         4957  0.0  0.4 388168  8860 ?        Sl   02:34   0:00 /usr/libexec/
gvfsd-dnssd --spawner :1.24 /org/gtk/gvfs/exec_spaw/2
root         6321  0.0  0.0      0     0 ?        I    02:37   0:00 [kworker/0:0-
ata_sff]
kali         9284  0.2  2.8 509760 58384 ?        Sl   02:43   0:02 /usr/bin/qter
minal
kali         9295  0.0  0.9 556900 19944 ?        Ssl  02:43   0:00 /usr/libexec/
xdg-desktop-portal
kali         9300  0.0  0.3 308744  6528 ?        Ssl  02:43   0:00 /usr/libexec/
xdg-permission-store
kali         9308  0.0  0.3 614484  7116 ?        Ssl  02:43   0:00 /usr/libexec/
xdg-document-portal
root         9322  0.0  0.0   2584  1840 ?        Ss   02:43   0:00 fusermount3 -
o rw,nosuid,nodev,fsname=portal,auto_unmount,subtype=portal -- /run/user/1000/do
c
kali         9329  0.0  0.9 406584 19248 ?        Ssl  02:43   0:00 /usr/libexec/
xdg-desktop-portal-gtk
kali         9346  0.2  0.3  10420  6368 pts/0    Ss   02:43   0:02 /usr/bin/zsh
root        16337  0.0  0.0      0     0 ?        I    02:57   0:00 [kworker/u4:0
]
root        16468  0.0  0.0      0     0 ?        I    02:58   0:00 [kworker/0:1-
ata_sff]
kali        17144  0.0  0.0   2680  1704 pts/0    S+   02:59   0:00 sh -c ps -aux
kali        17635 42.8  0.2  10176  4984 pts/0    R+   03:00   0:00 ps -aux
msfadmin@metasploitable:~$
```

Scansione con Nmap:
Viene eseguita una scansione completa dei servizi attivi sulla macchina target (Meta) utilizzando **Nmap**. La scansione permette di identificare le porte aperte, i protocolli in ascolto e i servizi associati.

Da kali, col comando **nmap 192.168.50.101** è stata effettuata una scansione di base delle porte.

```
┌──(nemi㉿kali)-[~]
└─ nmap 192.168.50.101
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-08 03:10 CEST
Nmap scan report for 192.168.50.101
Host is up (0.00073s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
23/tcp   open  telnet
25/tcp   open  smtp
53/tcp   open  domain
80/tcp   open  http
111/tcp  open  rpcbind
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
512/tcp  open  exec
513/tcp  open  login
514/tcp  open  shell
1099/tcp open  rmiregistry
1524/tcp open  ingreslock
2049/tcp open  nfs
2121/tcp open  ccproxy-ftp
3306/tcp open  mysql
5432/tcp open  postgresql
5900/tcp open  vnc
6000/tcp open  X11
6667/tcp open  irc
8009/tcp open  ajp13
8180/tcp open  unknown
MAC Address: 08:00:27:09:7A:98 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 13.50 seconds
```

Con **nmap -A 192.168.50.101** sono state rilevate informazioni aggiuntive, come versione dei servizi, sistemi operativo e script di rilevamento vulnerabilità.

```
┌──(nemi㉿kali)-[~]
└─ nmap -A  192.168.50.101
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-08 03:14 CEST
Nmap scan report for 192.168.50.101
Host is up (0.00068s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE     VERSION
21/tcp   open  ftp         vsftpd 2.3.4
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
| ftp-syst:
|   STAT:
| FTP server status:
|      Connected to 192.168.50.100
|      Logged in as ftp
|      TYPE: ASCII
|      No session bandwidth limit
|      Session timeout in seconds is 300
|      Control connection is plain text
|      Data connections will be plain text
|      vsFTPd 2.3.4 - secure, fast, stable
|_End of status
22/tcp   open  ssh         OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
| ssh-hostkey:
|   1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_  2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp   open  telnet      Linux telnetd
25/tcp   open  smtp        Postfix smtpd
|_ssl-date: 2025-09-08T01:15:13+00:00; 0s from scanner time.
|_smtp-commands: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCOD
, 8BITMIME, DSN
| ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is
 such thing outside US/countryName=XX
| Not valid before: 2010-03-17T14:07:45
|_Not valid after:  2010-04-16T14:07:45
| sslv2:
|   SSLv2 supported
|   ciphers:
|     SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
|     SSL2_DES_192_EDE3_CBC_WITH_MD5

|     SSL2_DES_192_EDE3_CBC_WITH_MD5
|     SSL2_RC2_128_CBC_WITH_MD5
|     SSL2_RC4_128_EXPORT40_WITH_MD5
|     SSL2_RC4_128_WITH_MD5
|_    SSL2_DES_64_CBC_WITH_MD5
53/tcp   open  domain      ISC BIND 9.4.2
| dns-nsid:
|_  bind.version: 9.4.2
80/tcp   open  http        Apache httpd 2.2.8 ((Ubuntu) DAV/2)
|_http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
|_http-title: Metasploitable2 - Linux
111/tcp  open  rpcbind     2 (RPC #100000)
| rpcinfo:
|   program version    port/proto  service
|   100000  2            111/tcp   rpcbind
|   100000  2            111/udp   rpcbind
|   100003  2,3,4       2049/tcp   nfs
|   100003  2,3,4       2049/udp   nfs
|   100005  1,2,3      33911/udp   mountd
|   100005  1,2,3      52691/tcp   mountd
|   100021  1,3,4      43407/udp   nlockmgr
|   100021  1,3,4      59065/tcp   nlockmgr
|   100024  1          46925/tcp   status
|_  100024  1          48442/udp   status
139/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp  open  netbios-ssn Samba smbd 3.0.20-Debian (workgroup: WORKGROUP)
512/tcp  open  exec        netkit-rsh rexecd
513/tcp  open  login       OpenBSD or Solaris rlogind
514/tcp  open  shell       Netkit rshd
1099/tcp open  java-rmi    GNU Classpath grmiregistry
1524/tcp open  bindshell   Metasploitable root shell
2049/tcp open  nfs         2-4 (RPC #100003)
2121/tcp open  ftp         ProFTPD 1.3.1
3306/tcp open  mysql       MySQL 5.0.51a-3ubuntu5
| mysql-info:
|   Protocol: 10
|   Version: 5.0.51a-3ubuntu5
|   Thread ID: 8
|   Capabilities flags: 43564
|   Some Capabilities: SwitchToSSLAfterHandshake, Support41Auth, SupportsTransactions, SupportsCompression, Spe
```

```
s41ProtocolNew, ConnectWithDatabase, LongColumnFlag
|   Status: Autocommit
|_  Salt: h-EQIXa^4)mkQ4Opg6tg
5432/tcp open  postgresql  PostgreSQL DB 8.3.0 - 8.3.7
|_ssl-date: 2025-09-08T01:15:13+00:00; 0s from scanner time.
| ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is
 such thing outside US/countryName=XX
| Not valid before: 2010-03-17T14:07:45
|_Not valid after:  2010-04-16T14:07:45
5900/tcp open  vnc         VNC (protocol 3.3)
| vnc-info:
|   Protocol version: 3.3
|   Security types:
|_    VNC Authentication (2)
6000/tcp open  X11         (access denied)
6667/tcp open  irc         UnrealIRCd
8009/tcp open  ajp13       Apache Jserv (Protocol v1.3)
|_ajp-methods: Failed to get a valid response for the OPTION request
8180/tcp open  http        Apache Tomcat/Coyote JSP engine 1.1
|_http-favicon: Apache Tomcat
|_http-title: Apache Tomcat/5.5
|_http-server-header: Apache-Coyote/1.1
MAC Address: 08:00:27:09:7A:98 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop
Service Info: Hosts:  metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:l
ux_kernel

Host script results:
| smb-os-discovery:
|   OS: Unix (Samba 3.0.20-Debian)
|   Computer name: metasploitable
|   NetBIOS computer name:
|   Domain name: localdomain
|   FQDN: metasploitable.localdomain
|_  System time: 2025-09-07T21:15:05-04:00
8009/tcp open  ajp13       Apache Jserv (Protocol v1.3)
|_ajp-methods: Failed to get a valid response for the OPTION request
8180/tcp open  http        Apache Tomcat/Coyote JSP engine 1.1
|_http-favicon: Apache Tomcat
|_http-title: Apache Tomcat/5.5
|_http-server-header: Apache-Coyote/1.1
MAC Address: 08:00:27:09:7A:98 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop
Service Info: Hosts:  metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:l
ux_kernel

Host script results:
| smb-os-discovery:
|   OS: Unix (Samba 3.0.20-Debian)
|   Computer name: metasploitable
|   NetBIOS computer name:
|   Domain name: localdomain
|   FQDN: metasploitable.localdomain
|_  System time: 2025-09-07T21:15:05-04:00
|_smb2-time: Protocol negotiation failed (SMB2)
|_nbstat: NetBIOS name: METASPLOITABLE, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)
|_clock-skew: mean: 59m59s, deviation: 2h00m00s, median: 0s
| smb-security-mode:
|   account_used: guest
|   authentication_level: user
|   challenge_response: supported
|_  message_signing: disabled (dangerous, but default)

TRACEROUTE
HOP RTT     ADDRESS
1   0.68 ms 192.168.50.101

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 37.56 seconds
```

Ex Facoltativo - Analisi del traffico di rete:

Con **Wireshark** sono stati catturati e analizzati i pacchetti scambiati tra Kali e Metasploitable durante le scansioni **Nmap**.

```
┌──(nemi㉿kali)-[~]
└─💜 nmap -sS 192.168.50.101
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-08 03:12 CEST
Nmap scan report for 192.168.50.101
Host is up (0.00061s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
23/tcp   open  telnet
25/tcp   open  smtp
53/tcp   open  domain
80/tcp   open  http
111/tcp  open  rpcbind
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
512/tcp  open  exec
513/tcp  open  login
514/tcp  open  shell
1099/tcp open  rmiregistry
1524/tcp open  ingreslock
2049/tcp open  nfs
2121/tcp open  ccproxy-ftp
3306/tcp open  mysql
5432/tcp open  postgresql
5900/tcp open  vnc
6000/tcp open  X11
6667/tcp open  irc
8009/tcp open  ajp13
8180/tcp open  unknown
MAC Address: 08:00:27:09:7A:98 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 13.36 seconds
```

Con **nmap -sS 192.168.50.101** è stata eseguita una scansione **SYN** per individuare le porte aperte senza completare il 3-way handshake, riducendo il rumore generato sulla rete e risultando più discreto.

```
No. Time        Source       Destination  Protocol Length Info
 0.104821630  192.168.5…  192.168.5…  TCP  60 1233 → 41814 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
 0.104821749  192.168.5…  192.168.5…  TCP  60 3703 → 52298 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
 0.104845460  192.168.5…  192.168.5…  TCP  74 33032 → 55600 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1525515396 TSecr=0 WS=128
 0.104899017  192.168.5…  192.168.5…  TCP  60 1076 → 34354 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
 0.104931588  192.168.5…  192.168.5…  TCP  74 55224 → 1095 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1525515396 TSecr=0 WS=128
 0.104987218  192.168.5…  192.168.5…  TCP  60 55600 → 33032 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
 0.105009679  192.168.5…  192.168.5…  TCP  74 60408 → 6346 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1525515396 TSecr=0 WS=128
 0.105069957  192.168.5…  192.168.5…  TCP  60 1095 → 55224 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
 0.105086525  192.168.5…  192.168.5…  TCP  74 57920 → 2382 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1525515396 TSecr=0 WS=128
 0.105114792  192.168.5…  192.168.5…  TCP  74 37506 → 1166 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1525515396 TSecr=0 WS=128
 0.105170886  192.168.5…  192.168.5…  TCP  60 6346 → 60408 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
 0.105235834  192.168.5…  192.168.5…  TCP  60 2382 → 57920 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
 0.105273760  192.168.5…  192.168.5…  TCP  74 46192 → 10002 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1525515397 TSecr=0 WS=128
 0.105345670  192.168.5…  192.168.5…  TCP  60 1166 → 37506 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
 0.105412935  192.168.5…  192.168.5…  TCP  60 10002 → 46192 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
 0.105433085  192.168.5…  192.168.5…  TCP  74 41614 → 49400 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1525515397 TSecr=0 WS=128
 0.105462837  192.168.5…  192.168.5…  TCP  74 52368 → 1084 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1525515397 TSecr=0 WS=128
 0.105483128  192.168.5…  192.168.5…  TCP  74 39920 → 4279 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1525515397 TSecr=0 WS=128
 0.105549515  192.168.5…  192.168.5…  TCP  74 49748 → 2107 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1525515397 TSecr=0 WS=128
 0.105670079  192.168.5…  192.168.5…  TCP  60 49400 → 41614 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
 0.105670235  192.168.5…  192.168.5…  TCP  60 1084 → 52368 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
 0.105670299  192.168.5…  192.168.5…  TCP  60 4279 → 39920 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
 0.105773791  192.168.5…  192.168.5…  TCP  60 2107 → 49748 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0

› Frame 2024: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface eth0, id 0
› Ethernet II, Src: PCSSystemtec_09:7a:98 (08:00:27:09:7a:98), Dst: PCSSystemtec_d1:f8:5d (08:00:
› Internet Protocol Version 4, Src: 192.168.50.101, Dst: 192.168.50.100
› Transmission Control Protocol, Src Port: 1233, Dst Port: 41814, Seq: 1, Ack: 1, Len: 0

0000  08 00 27 d1 f8 5d 08 00  27 09 7a 98 08 00 45 00   ··'··]·· '·z···E·
0010  00 28 00 00 40 00 40 06  54 b6 c0 a8 32 65 c0 a8   ·(··@·@· T···2e··
0020  32 64 04 d1 a3 56 00 00  00 00 32 37 e8 03 50 14   2d···V·· ··27··P·
0030  00 00 07 54 00 00 00 00  00 00 00 00               ···T···· ····
```

Nel traffico catturato con **Wireshark** si osservano pacchetti **SYN** e **SYN-ACK**, senza l’invio del pacchetto **ACK** finale. Questo conferma che la scansione **SYN** si limita a iniziare il 3-way handshake senza completarlo.

```
┌──(nemi㉿kali)-[~]
└─💜 nmap -sT  192.168.50.101
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-08 03:27 CEST
Nmap scan report for 192.168.50.101
Host is up (0.00069s latency).
Not shown: 977 closed tcp ports (conn-refused)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
23/tcp   open  telnet
25/tcp   open  smtp
53/tcp   open  domain
80/tcp   open  http
111/tcp  open  rpcbind
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
512/tcp  open  exec
513/tcp  open  login
514/tcp  open  shell
1099/tcp open  rmiregistry
1524/tcp open  ingreslock
2049/tcp open  nfs
2121/tcp open  ccproxy-ftp
3306/tcp open  mysql
5432/tcp open  postgresql
5900/tcp open  vnc
6000/tcp open  X11
6667/tcp open  irc
8009/tcp open  ajp13
8180/tcp open  unknown
MAC Address: 08:00:27:09:7A:98 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 13.32 seconds
```

Il comando **nmap -sT 192.168.50.101** completa il three-way handshake su ciascuna porta target, ottenendo informazioni dettagliate sulle porte aperte e sui servizi, ma generando più traffico di rete.

```
No. Time         Source        Destination   Protocol Length Info
    0.104821630  192.168.5…    192.168.5…    TCP      60 1233 → 41814 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
    0.104821749  192.168.5…    192.168.5…    TCP      60 3703 → 52298 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
    0.104845460  192.168.5…    192.168.5…    TCP      74 33032 → 55600 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1525515396 TSecr=0 WS=128
    0.104899017  192.168.5…    192.168.5…    TCP      60 1076 → 34354 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
    0.104931588  192.168.5…    192.168.5…    TCP      74 55224 → 1095 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1525515396 TSecr=0 WS=128
    0.104987218  192.168.5…    192.168.5…    TCP      60 55600 → 33032 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
    0.105009679  192.168.5…    192.168.5…    TCP      74 60408 → 6346 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1525515396 TSecr=0 WS=128
    0.105069957  192.168.5…    192.168.5…    TCP      60 1095 → 55224 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
    0.105086525  192.168.5…    192.168.5…    TCP      74 57920 → 2382 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1525515396 TSecr=0 WS=128
    0.105114792  192.168.5…    192.168.5…    TCP      74 37506 → 1166 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1525515396 TSecr=0 WS=128
    0.105170886  192.168.5…    192.168.5…    TCP      60 6346 → 60408 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
    0.105235834  192.168.5…    192.168.5…    TCP      60 2382 → 57920 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
    0.105273760  192.168.5…    192.168.5…    TCP      74 46192 → 10002 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1525515397 TSecr=0 WS=128
    0.105345670  192.168.5…    192.168.5…    TCP      60 1166 → 37506 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
    0.105412935  192.168.5…    192.168.5…    TCP      60 10002 → 46192 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
    0.105433085  192.168.5…    192.168.5…    TCP      74 41614 → 49400 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1525515397 TSecr=0 WS=128
    0.105462837  192.168.5…    192.168.5…    TCP      74 52368 → 1084 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1525515397 TSecr=0 WS=128
    0.105483128  192.168.5…    192.168.5…    TCP      74 39920 → 4279 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1525515397 TSecr=0 WS=128
    0.105549515  192.168.5…    192.168.5…    TCP      74 49748 → 2107 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1525515397 TSecr=0 WS=128
    0.105670079  192.168.5…    192.168.5…    TCP      60 49400 → 41614 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
    0.105670235  192.168.5…    192.168.5…    TCP      60 1084 → 52368 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
    0.105670299  192.168.5…    192.168.5…    TCP      60 4279 → 39920 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
    0.105773791  192.168.5…    192.168.5…    TCP      60 2107 → 49748 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0

› Frame 2024: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface eth0, id 0    0000  08 00 27 d1 f8 5d 08 00  27 09 7a 98 08 00 45 00   ··'··]·· '·z···E·
› Ethernet II, Src: PCSSystemtec_09:7a:98 (08:00:27:09:7a:98), Dst: PCSSystemtec_d1:f8:5d (08:00:  0010  00 28 00 00 40 00 40 06  54 b6 c0 a8 32 65 c0 a8   ·(··@·@· T···2e··
› Internet Protocol Version 4, Src: 192.168.50.101, Dst: 192.168.50.100                            0020  32 64 04 d1 a3 56 00 00  00 00 32 37 e8 03 50 14   2d···V·· ··27··P·
› Transmission Control Protocol, Src Port: 1233, Dst Port: 41814, Seq: 1, Ack: 1, Len: 0           0030  00 00 07 54 00 00 00 00  00 00 00 00               ···T···· ····
```

Nel traffico catturato con **Wireshark** si osserva il completamento del 3-way handshake **TCP (SYN, SYN-ACK, ACK)**, che stabilisce una connessione reale con la porta target. Questo conferma il comportamento della scansione **TCP completa (-sT).**